
UNIT 2 UNDERSTANDING DATA AND ITS TYPE

Structure

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2.0 INTRODUCTION

Data is the essential foundation for every data science effort a systematically recorded information, along with its context, is referred to as data. All our actions can be seen in the form of data. For instance, you watch television in your home. Data scientists will see it as:

- Proportion of time you allocated to watching news, films, sports, etc.
- TV channel you watched.
- Start and end time of watching the television.
- Posture while watching television and similar activities.

In essence, data science facilitates the datafication of several facets of human life. This data may subsequently be utilised to derive insights, formulate forecasts, and enhance several facets of life and business. Consequently, datafication facilitates the digital representation of the physical world and serves as the foundation of the fourth industrial revolution. Nonetheless, datafication results in a substantial volume of data. Diverse events can be recorded in many forms of data. Some are presented in tabular style, while others are in textual or video format. Data may be both quantitative and

qualitative. Some are quantified as absolute values; others are recorded as ranges. Some measurements are continuous, whereas others occur at discrete intervals. The quantity and amount of data will also vary depending on the application. This unit will cover various forms of data and levels / Stages of measurement. Regardless of the format of data, the data quality is paramount. This unit will also emphasise several metrics of data quality.

2.2 OBJECTIVES

On completion of this unit, you should be able to:

- To understand various data types and formats
- To understand measures of quality of data
- To understand levels of measurement of data

2.3 PROPERTIES OF DATA

Organizations have acknowledged the power of informed, data-backed decisions for their operations. The effectiveness of data-backed decision-making primarily depends on the quality of the data. Inferior input yields inferior output. Inadequate data quality might result in inferior decision-making and unnecessary expenditures or possible business losses. Evaluating and improving data quality is essential for the successful execution of data science projects and the overall performance of the organization. The following aspects of data are crucial in assessing data quality.

- **Completeness:** The degree to which a dataset contains all the essential information is called data completeness. Completeness of data ensures correct analysis, reliable insights and avoids misleading conclusions. Simplified and standard data entry process, well documented data collection protocol and data validation check are required to ensure data completeness. In addition, awareness sessions and training of concerned stakeholders will help.
- **Accuracy:** Accuracy is the degree to which collected data is free from errors and how closely it corresponds to its actual value. Accuracy helps to make decisions that are correct and strengthen stakeholders' trust. It also saves time and cost involved in decision projects. Checks for data errors, inconsistencies and inaccuracies are often made at the early stage of any data science project. We need to use reliable data sources to get accurate data. In addition, organizations must evolve internal standards for data collection and storing to ensure accuracy.
- **Consistency:** It relates to the uniform interpretation of analogous information throughout the diverse sources. In other words, the identical data point should possess the same value in every location. It guarantees data precision and cultivates organisational trust in data. Data integrity and effective decision-making rely on consistency of the data, which ensures that the data will remain consistent, reliable, and accurate across multiple platforms and systems. Inconsistent data can lead to misinterpretations, mistakes, and a lack of trust in the information.

- **Timeliness:** Timeliness refers to how current and up-to-date the information is. Data should be readily available whenever it's needed and undergo real-time updates. In an age of ever-evolving information, timeliness is seen as an essential characteristic of data quality. Organizations can get a competitive advantage by implementing efficient data collection, storage, and updating processes, enabling informed decision-making based on the latest information available.
- **Uniqueness:** Uniqueness ensures that each data point in a dataset is captured once, eliminating duplication and overlap. Uniqueness boosts data correctness and reliability, enabling accurate analysis and decision-making. Data cleansing and deduplication provide unique records, enhancing data governance and compliance. Thus, data uniqueness is essential to data management and organizational success.
- **Validity and Integrity:** Data validity focuses on whether data adheres to specific predefined rules and criteria, confirming its accuracy and relevance for its intended purpose. Data validity is a key component of maintaining data integrity. Better decision-making and operational efficiency can be achieved when organizations maintain high data validity, which enhances the credibility and precision of their data.
- **Reliability:** Reliability refers to data accuracy, across the several sources and systems. Reliable data guarantees that the information utilized for analysis and decision-making adheres to the allowed limits for the specific task or question. Data reliability is enhanced by the implementation of validation criteria and the maintenance of audit trails. Reliable data provides a solid foundation of accurate and trustworthy information.
- **Relevancy:** Relevance to the degree to which collected data is necessary and helpful in achieving specific goals. Data gathering efforts could be fruitless and ineffective if the data is not relevant. Data that is relevant is the result of procedures that establish objectives, assess the data's requirement, and synchronise it with applications.

☞ Check Your Progress 1

1. Define any 5 characteristics of good quality data.
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2. Why is timely updating of information important for any organization?
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2.4 DATAFICATION

Data science sees every activity in the form of data. It essentially converts everything we do into data. This process of converting various aspects of our lives into data is called datafication. Datafication is then used to draw insights, make predictions, and improve various aspects of life and business. Following table will further describe the datafication concept with the help of two examples.

	Use case 1	Use case 2
Real world activity	You walk daily to keep yourself healthy	A machine is producing jobs in a manufacturing plant
Datafication	• Number of Steps • Duration of walk • Heart rate record • Footwear used • Track type and condition • Person's age • Person's medical history	• Start and stop time • Jobs characteristics • Production stop reasons • Operators' details • Energy consumption • Quality of the jobs produced • Process parameters • Machine condition parameters
Potential benefits of datafication	• understand your health and fitness levels • alert you to potential health issues • better, more personalized care	• Predict potential failure • Optimize process parameters • Reduce energy consumption • Operator assessment and allocation • Reduce defective products

☞ Check Your Progress 2

1. Do datafication of your television watching activity or mobile phone usage activity.

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2.5 DATA CATEGORIES

Datafication provides data. Two types of data are possible: quantitative and qualitative. It's important to know the difference between the two because they need different ways to collect and analyse data.

Quantitative data implies the data which can be measured or counted and expressed in the form of numerical values. Accordingly, it gives us

information about “how many,” “how much,” or “how often”.

For example, following information can be captured in the form of quantitative data.

- How many students opt for data science courses in the university in the last 5 years?
- How much weight loss did a person achieve in the last month?
- How often do you buy products from online shopping apps?

Qualitative data is not quantifiable or countable. It is characterised by descriptive language rather than numerical figures. Consequently, it provides answers to "Why?" or "How?" questions.

For example, following information can be captured in the form of qualitative data.

- Why did you not find service offered by a vendor up to the mark?
- How does a user feel about a particular product?

You are not just looking at numbers in these situations; you are also asking the person to explain in words why they did something or how they feel. Qualitative data are normally subjective, and two persons may have different opinions for the same thing. Quantitative data is normally fixed and universal and does not change.

Quantitative data is generated by measuring and counting things. Qualitative data are generated mostly through observations and interviews.

Statistical analysis and mathematical computations are employed to evaluate quantitative data, whereas qualitative data is typically examined by categorizing it into meaningful themes or groups.

Check Your Progress 3

1. Go for an evening walk and record the number of steps, start and end time of walk. Comment if this information is qualitative or quantitative.

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2. Talk to your friends and check their experience with the mobile phone they are using currently. Comment if you get qualitative or quantitative data.

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2.6 DATA FORMAT

The data format defines the structure of data within a system be it database/filebase system, imparting relevance to the processed data i.e. information. Sources of the data may exist in various formats. The following three formats are often utilized.

- Structured/Formatted data
- Unstructured/Unformatted data
- Semi-structured /Semi-formatted data

2.6.1 Structured/Formatted Data

It relates to the organized data that is data formatted into tables and encompasses discrete data types, including numerical values, short text, and dates. Structured data is typically characterised by rows and columns, with columns denoting various attributes/elements such as emp_name, emp_address, and emp_phone number, each field possessing a certain type, including various data types viz. integers or floating numbers, Strings or characters, also Boolean type data. Rows denote individuals' data i.e. record, that is to fill each attribute/element represented by a column by its corresponding value.

Examples of structured data with commonly used options to store structured data are given below.

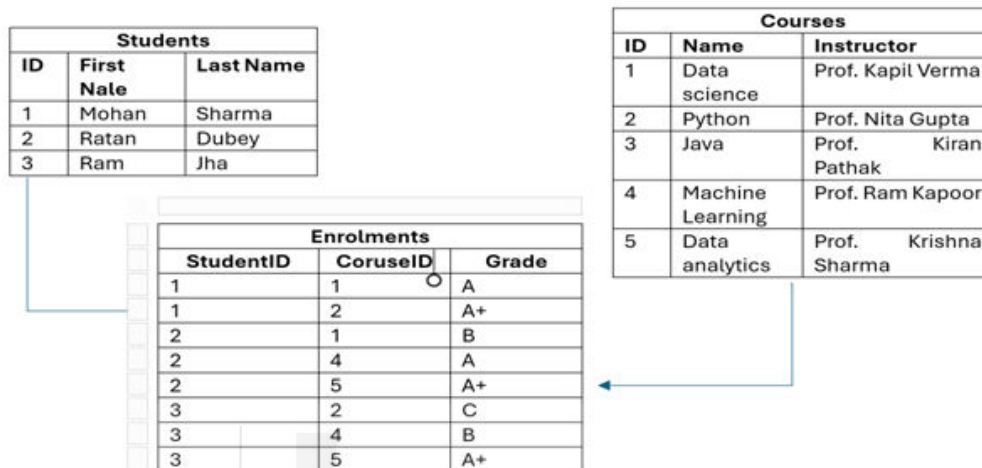
- **Spreadsheets:** Spreadsheets are normally used to store data in tables with relationships between rows and columns. Normally well suited for a small volume of structured data. An example is shown below.

Year wise admissions in various branches of engineering in various institutes

Institute code	Mechanical	Electrical	Computer Science	Mechanical	Mechanical	Electrical	Mechanical	Electrical	Computer Science
101	50	50	50	48	50	50	46	49	50
102	40	50	50	36	47	49	43	50	49
103	35	50	50	37	47	50	38	49	50
104	50	49	49	48	48	49	47	49	49
105	49	39	48	50	39	50	50	40	48
106	43	45	50	47	50	50	45	50	50
107	38	50	47	43	39	50	38	46	50
108	25	47	50	50	47	50	37	43	50
109	39	48	47	45	49	48	49	49	50
110	48	43	50	48	50	50	50	43	50

- **Relational databases (RDB):** It is a form of structured storage of data that organizes and stores data in the form of a tables, i.e. rows & columns. Relational databases are designed to create connections among

data elements, facilitating data comprehension and analysis. Every row possesses a distinct identifier, or key, that connects tables. Relational databases utilize SQL (Structured Query Language) for the creation and management of databases. Relational databases are efficient at managing large amounts of structured data. A sample structure of a relational database with one example is shown below.



2.6.2 Unstructured/Unformatted data

Unstructured data consists of information that does not conform to a set data model and is not stored in a structured manner, which can be easily read by the machines. Unlike structured data, which is stored in relational databases with rows and columns, unstructured data is typically in formats like digital text, Graphics/images, Animation/videos, speeches/audio, media posts, emails, and sensor-driven data. Sometimes, numerical or textual data can be unstructured because modeling it as a table is inefficient. Imagine a temperature sensor in a factory: it continuously outputs temperature readings at various intervals, it would be considered unstructured data unless specifically formatted into a structured database with timestamp and temperature values as separate columns. Typically, unstructured data requires advanced tools from the field of (DL) Deep Learning/(ML) Machine Learning which specifically includes NLP i.e. Natural Language Processing (NLP) to extract insights.

Examples of unstructured data

- Text: Emails, text messages,
- Images: Photos in JPEG, GIF, or PNG formats
- Audio: Audio files in MP3, WAV, or FLAC formats
- Video: Video files in MP4, AVI, or MOV formats
- Healthcare data: MRIs, x-rays, CT scans, doctor's notes,
- Surveillance data: Video monitoring data
- Weather data: Weather data
- Internet of Things (IoT) data: sensor data from devices
- Social media posts: Posts on social media platforms

2.6.3 Semi-Structured/Semi-Formatted data

It represents the data type that doesn't follow the strict format / structure of traditional databases but still includes elements for data organization like markers or tags to separate data elements. It sits between structured and unstructured data, making it more flexible yet partially organized. Thus, semi-structured data are partially structured and simpler to process compared to unstructured data but not as straightforward as structured data.

- JSON and XML files
- NoSQL databases (e.g., MongoDB)
- Emails with headers (subject, sender, etc.)
- Sensor data with labels
- HTML code
- XML documents

HTML (HyperText Markup Language): On the web, HTML is the standard markup language used to produce and organise content. It offers the organisation and layout of text, images, links, and other material using a system of components and tags, therefore defining the framework for web pages. It organizes content using tags like <header>, <p>, and <div>. It connects pages through links using the <a> tag. It embeds images, videos, and audio files with tags like , <video>, and <audio>. There are various tags for the formatting and presentation of data on any web page, an example of THML code is given below.

```
<!DOCTYPE html>
<html>
<head>
  <title>Welcome to Our Website</title>
</head>
<body>
  <h1>Hello, Earth! </h1>
  <p> a simple HTML example.</p>
  <a href="https://www.example.com">Visit Example</a>
</body>
</html>
```

XML document: XML stands for eXtensible Markup Language, it is a markup language created for the storage and transmission of hierarchical data in a structured format which can be read by both humans and machine. It is extensively utilised for data interchange between systems, configuration files, and many applications requiring flexible data representation. XML is platform-agnostic and adheres to a hierarchical framework with nested elements. Each element is defined using start and end tags. XML is expandable, allowing for the definition of custom tags to meet specific data requirements.

Example:

```
<?xml version="1.0" encoding="UTF-8"?>
<library>
  <book>
    <title>Data Science</title>
    <author> IGNOU </author>
    <year>2025</year>
  </book>
  <book>
    <title>Data Analytics</title>
    <author>IGNOU</author>
    <year>2025</year>
  </book>
</library>
```

The <library> tag is the root element.

Each <book> element contains child elements like <title>, <author>, and <year> to describe a book.

Tags like <title> and <author> are custom tags created for the specific data.

JSON (JavaScript Object Notation): it is a data interchange standard which is lightweight in nature and it allows user to perform read / write operations to comprehend and produce. Its simplicity and adaptability for many programming languages enable its great use for data sharing in APIs and web apps.

JSON is based on a collection of key-value pairs which is quite similar to the concept of dictionary in Python and that of objects in JavaScript. It supports data types like strings, numbers, arrays, objects, booleans, and null.

```
{
  "employees":[
    {"firstName":"Mohan", "lastName":"Joshi"},
    {"firstName":"Aman", "lastName":"Sharma"},
    {"firstName":"Ramesh", "lastName":"Verma"}
  ]
}
```

CSV (Comma-Separated Values) file: it is commonly used for storing structured data. An example is given below.

Candidate, Age, Country, Education, Institute, Years of Experience,

Manoj, 34, India, PhD, IIT, 1

Suresh, 37, India, PhD, NIT, 0

Ramesh, 33, India, MTech, IIT, 5

Another similar way to store the structured data is to use the Tab-Separated Values (TSV) file. In the TSV file we use tab to separate the entry instead of comma.

Semi-Structured Vs Structured data:

Semi-structured and structured data differs in two key aspects: schema and data structure.

The first distinction is schema. Unlike structured data, which requires a predefined and fixed schema, semi-structured data does not rely on a prior schema definition. This lack of rigidity makes semi-structured data more adaptable, allowing it to evolve as new attributes are added over time.

The second difference is in data structure. Semi-structured data uses a hierarchical structure that can include nested information, whereas structured data is represented in a flat, tabular format. This hierarchical organization makes semi-structured data particularly well-suited for handling data from applications and internet-enabled devices

Semi-Structured Vs Unstructured data:

Unstructured/Unformatted data is raw data that lacks an established model or schema. In contrast, semi-structured data has identifiable and consistent markers that define distinct semantic elements, organizing the data into a hierarchy of records and fields.

Check Your Progress 4

1. Write major differences between structured and unstructured data.

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2. Give three examples of both structured and unstructured data.

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3. Choose the best possible format for the following data

- Customer account data
- Product photos and videos
- X-Rays and MRI scans
- Doctors' prescriptions
- Medical billing data
- Financial transaction

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2.7 PRIMARY DATA & SECONDARY DATA:

The **Primary data** relates to the firsthand data, collected primarily to facilitate any study. The sources of primary data include observations, questionnaires, interviews, and experiments.

- **Observations:** It involves observing and collecting events, patterns, phenomena or behaviors as they occur in their actual setting. Few examples of data acquisition/collection through observation are given below.
 - Observing customers' purchasing behaviour in the mall to improve layout.
 - Observing production workers' activities to improve line efficiency.
 - Observing support staff's behaviour in hospital for improving customers' experience.
- **Questionnaires:** These are structured or semi-structured sets of questions distributed to individuals to gather information. They can be controlled or regulated in person, or remotely through online forms or email. For example, many times restaurants collect feedback from customers in the form of a questionnaire. It includes questions such as- are you satisfied with the quality of food? How do you rate the quality of service? The answers could be in the form of options like-poor, average good, very good.
- **Interviews:** It involves direct interaction, where researchers ask respondents questions to collect in-depth qualitative or quantitative data. A company conducts interviews of some of its potential customers to understand their requirements related to the products offered by the company.
- **Experiments:** Researchers can collect data through experiments from the testbeds or from the actual system installed in the field. For example, in order to predict the reliability of a bearing, researchers can run some bearings using a test-rig in accelerated condition to capture information regarding time to failure and failure modes.

Secondary data: It relates to the information which is previously collected or recorded for the purposes, which may be different from the present research. The sources of secondary data include company reports, government of India reports, bulletins, etc.

- **Company Annual Reports:** It includes annual financial statements, market analysis reports, or internal audit summaries published by organizations. For example, various companies release their annual financial statements, providing an overview of their fiscal performance in a particular financial year. Various data analytics companies use secondary data for making predictions about share market and market growth in a particular sector.

- **Government Reports:** It includes official publications such as census data, economic surveys, and policy documents. For example, the (MoE) Ministry of Education, (GoI) Government of India, publishes an annual report titled National Institutional Ranking Framework (NIRF), which performs evaluation and ranking of educational institutions across the Nation(India), based on well-defined parameters. If we use such parameters and ranking for our study, then this becomes secondary data for our study.
- **Data from previous studies or experiments:** Data from previous studies or experiments can be used as secondary data for any new study. For example, some researchers conducted some experiments and collected data for reliability estimation of a particular bearing. The same can be used as secondary data by another researcher in future for some other study.

☞ Check Your Progress 5

1. Design a questionnaire and collect primary data for the feedback about a movie from your friends' group.

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2. Using some of the published census data of the government of India, draw some conclusion. Comment if the data used by you is primary or secondary data

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2.8 CATEGORICAL DATA & NUMERICAL DATA

It relates to the data which is partitioned into separate categories or groups. These categories are typically labels or names that describe qualities or characteristics and do not have a natural numerical ordering.

Examples:

- Machine type: Lathe, Milling, CNC
- Product categories: Automotive, Aerospace, Electronics
- Gender: Male, Female, Other
- Type of family: Nuclear, joint

These types of data can be further divided as nominal or ordinal data based on the level of measurement. The categories or groups without any inherent

order (e.g., color, gender, choice,) is called nominal data. Whereas the categories or the groups with a meaningful order but no fixed numerical difference (e.g., rating scales like Poor, Average, Good) is called ordinal data.

The Numerical data comprises the data in the form of numbers, and it is used to represent the measurable quantities. This numerical data can be mathematically analyzed and is often used for statistical computations.

Examples:

- Number of units produced per hour: 100, 200, 300
- Height: 120 cm, 140 cm, 150 cm
- Cycle time: 30.5 sec, 40.2 sec, 50 sec

It can be seen from above examples that the numerical data can be discrete data (whole number or counts) or continuous data (values within a range, including decimals).

Check Your Progress 6

1. Collect a few data by measuring various events or objects around you. Classify your measurement as Categorical data and numerical data.

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2. Use a thermometer and measure your body at various times in a day. Comment if the data collected by you is numerical or categorical data.

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2.9 CROSS-SECTIONAL DATA AND TIME SERIES DATA

It is the information gathered across several elements/attributes — such as households, firms, or regions of any individual —at a particular point /instance of time or over a limited period of time. It provides a snapshot of variables and their relationships at that specific moment. Unlike time-series data, which tracks changes over time, cross-sectional data focuses on variations across entities at one time.

Key Characteristics:

1. **One-time observation:** Data is collected at a particular point /instance of time.
2. **Multiple subjects:** Observations/Recordings are made across various entities.
3. **Static analysis:** Focuses on variations between subjects rather than changes over time.

Example:

Suppose a study aims to analyse the number of calls received at CM helpline numbers from various districts of the state during a particular month.

Day	Region 1	Region 2	Region 3	...	Region n
1	20	25	35		45
2	20	43	25		35
3	10	23	14		24
.					
.					
.					
30	4	35	24		14

Cross-sectional data is frequently used in fields like sociology, economics, and healthcare to identify patterns, correlations, or comparisons across a population.

Time series data: It relates to a sequence of recordings / observations performed at regular intervals of time, which captures how a specific variable or phenomenon varies over a time period. Examples include stock market, product sales, hourly readings of temperature, or yearly growth rate of GDP .

Key characteristics of time series data include:

1. **Temporal Order:** Observations are inherently ordered by time, making the sequence crucial for analysis.
2. **Trends:** Long-term increases or decreases in the data over time.
3. **Seasonality:** Regular patterns or cycles that repeat at consistent intervals or gaps, like monthly or quarterly or half yearly or annually.
4. **Stationarity:** A property where statistical properties like central tendency of data, and its variation, do not change over time.
5. **Noise:** Random variations or irregularities superimposed on the data.

Time series data is widely used for forecasting, anomaly detection, and understanding underlying processes in various fields such as finance, meteorology, and economics. Analytical methods include statistical techniques like Exponential Smoothing, Autoregressive Integrated Moving Average (ARIMA), and machine learning models for complex patterns as well.

☞ Check Your Progress 7

1. Collect past weather records of your city using a smart phone weather app. Comment if the data collected by you is time series data or cross-sectional data.

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2. Make a data table to collect the following information from your family members- age, height, favorite food and favorite new channel. Comment if the data collected by you is time series data or cross-sectional data.

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2.10 DISCRETE DATA AND CONTINUOUS DATA

It represents the type of Data which takes specific or countable values, often integers, and cannot be subdivided meaningfully.

Examples:

- Number of defective items in a batch.
- Number of machines in a production line.

Continuous Data: Data that can take any value within a range and can be subdivided infinitely.

Examples:

- Dimensions of a machined component (e.g., length in millimeters).
- Pressure in a hydraulic system (e.g., in pascals).

Check Your Progress 8

1. Find at least 3 discrete and continuous data examples around you.

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2.11 LEVEL OF MEASUREMENT OF DATA: NOMINAL, ORDINAL, RATIO AND INTERVAL DATA SCALES, BIG DATA

Level of measurement, sometimes referred to as scale of measurement, is a classification of data depending on the type of information contained in the assigned values for variables. There are four levels of measurement.

- a) **Nominal level:** Nominal data represents the most basic level/stage of measurement. It is used for labelling or categorizing variables without any inherent order or quantitative value. The Nominal data categories are mutually exclusive, meaning each data point can belong to only one category, and there is no sense of ranking among them. This type of data is qualitative in nature. Examples are blood type (A, B, AB, O), marital status (single, married, divorced) and gender (male, female, other). While we can count how many individuals fall into each category, we cannot make any meaningful comparisons beyond that.

In short, the nominal level of measurement uses numbers to categories or classify the variables. These numbers do not have any order or hierarchy to them. Also, these numbers do not convey any numerical value or hierarchy.

For example, researchers collect data about pets.

- 1 Bird
- 2 Fish
- 3 Cat
- 4 Dog
- 5 Other

Such data is normally analyzed by frequency distribution.

- b) Ordinal level:** Ordinal data goes one step beyond nominal data by incorporating order among the categories. It allows for ranking or positioning, but the intervals/gaps between the ranks are not necessarily equal or meaningful. It means that although we can say one value is higher or lower than another, we cannot determine by how much. Ordinal data is still qualitative, although it is often treated as quasi-quantitative for practical analysis. Common examples include Satisfaction levels (very satisfied, satisfied, neutral, unsatisfied, very unsatisfied), education levels (Doctorate, Masters, Bachelors, high School), and rankings in a competition (1st, 2nd, 3rd). These examples demonstrate that we can understand relative position, but not the exact magnitude of difference between them.

In brief, the ordinal level of measurement numbers are used to represent rank or order, indicating relative positions. However, the relative magnitude or difference between the numbers is not meaningful.

Examples:

- 1 for president
- 2 for vice president
- 3 for plant manager
- 4 for supervisor

- c) Interval level:** Interval data introduces the concept of measurable and equal intervals between values. It is numeric and ordered, and the differences between values are meaningful. You can add and subtract interval data, but because it doesn't have a real zero point, comparing values as ratios doesn't work. For instance, with the degree of hotness measured in Celsius or Fahrenheit, 0 does not indicate the absence of hotness i.e. temperature—it is just an alternative point on the scale. Other examples include IQ scores and calendar dates (e.g., the year 2000 vs. 2020). It can be said that the temperature difference between 20°C and 10°C is the same as the temperature difference between 30°C and 20°C, and because the scale lacks an absolute zero, it cannot be said that 20°C is twice as hot as 10°C

So, we learned that Interval level measurement is a kind of data classification where the difference between values is meaningful and consistent, but there is no true zero point. This means the data can represent relative differences but not absolute quantities.

Examples of Interval Data are say, Degree of hotness in °C or °F, Time on a 12-hour clock

- d) **Ratio Level:** Ratio data is the highest or apex level/stage of measurement and covers all the properties of interval data, with the addition of a meaningful, absolute zero. It allows not only for differences and ordering, but also for meaningful ratios. Ratio data is quantitative and supports all mathematical operations, including multiplication and division. Examples like body weight (e.g., 0 kg, 5 kg, 10 kg), height (e.g., 0 cm, 150 cm, 180 cm), age (e.g., 0 years, 20 years), and income (e.g., \$0, \$30,000, \$60,000). Because ratio data has a true zero, we can say that 20 kg is twice as heavy as 10 kg or that a person earning \$90,000 earns six times as much as someone earning \$15,000.

The above-mentioned examples represent an interval level of data classification, where numerical values have meaningful magnitude and the differences between them are comparable. Additionally, the scale includes an absolute, natural zero point, signifying a true absence of the measured attribute.

Examples:

- Height
- Weight
- Volume
- Distance

2.11.2 Big data

Big Data describes exceptionally large and complex datasets that grow at an exponential rate. These data collections are so vast and dynamic that conventional data processing systems struggle to store, manage, or analyze them effectively.

Big Data is often defined by three key characteristics, commonly known as the 3Vs:

- **Volume** – Refers to the massive amount of data generated from various sources, often measured in terabytes, petabytes, or even exabytes.
- **Velocity** – Describes the high speed at which data is created, transmitted, and processed—often in real-time or near-real-time scenarios.
- **Variety** – Indicates the wide range of data types, including: Structured data like relational databases, Semi-structured data such as JSON or XML files, and Unstructured data like videos, images, emails, and social media content.

These datasets come from diverse origins, including social networking platforms, Internet of Things (IoT) devices, online transactions, industrial

sensors, and more. To handle and extract insights from such data, modern technologies and frameworks are used—such as Hadoop, Apache Spark, and cloud-based infrastructures. These are often paired with tools from machine learning, data mining, and analytics to derive valuable information.

Big Data has transformative applications across multiple sectors:

- In healthcare, it supports predictive diagnostics and personalized treatment.
- In finance, it helps detect fraudulent activities and manage risk.
- In retail, it enables customized marketing and enhances customer experience.
- In urban development, it contributes to the creation of smart cities through efficient planning and resource management.

By leveraging Big Data, organizations can make data-driven decisions, improve operational efficiency, and fuel innovation.

Check Your Progress 9

1. Choose the primary characteristics of big data among the following
 - (a) Value
 - (b) Variety
 - (c) Volume
 - (d) All of the above
2. What is Hadoop?
 - (a) A programming language for big data analytics
 - (b) A distributed file system for storing big data
 - (c) A framework for distributed storage and processing of big data
 - (d) A database management system for big data

2.12 SUMMARY

When you know what kind of data you have, it is easier to choose the right storage solutions, analysis methodologies, and tools to get useful insights. Every kind of data presents its own set of obstacles and opportunities when it comes to processing and interpretation. This chapter provided the groundwork for comprehending how to efficiently handle various types of data. This course examined a variety of data kinds and forms. Additionally, the quality of the data is discussed. Students can gather several types of data from their everyday activities and categorise it according to different formats and levels of measurement. Students are urged to form groups and talk about the data they have gathered.

2.13 SOLUTIONS/ANSWERS

☞ **Check Your Progress 1**

1. Refer section 2.3
2. Refer section 2.3

☞ **Check Your Progress 2**

1. Refer section 2.4

☞ **Check Your Progress 3**

1. Refer section 2.5
2. Refer section 2.5

☞ **Check Your Progress 4**

1. Refer section 2.6
2. Refer section 2.6
3. Refer section 2.6

☞ **Check Your Progress 5**

4. Refer section 2.7
5. Refer section 2.7

☞ **Check Your Progress 6**

1. Refer section 2.8
2. Refer section 2.8

☞ **Check Your Progress 7**

1. Refer section 2.9
2. Refer section 2.9

☞ **Check Your Progress 8**

1. Refer section 2.10

☞ **Check Your Progress 9**

1. (d)
2. (c)

2.14 FURTHER READINGS

1. Data Science from Scratch: First principles with Python, Joel Grus, 2nd Edition, O'REILLY, Shroff Publishers & Distributors Pvt. Ltd., 2021
2. Data Science for Beginners, Andrew Park, 1st Edition, Packt Publishing, 2020